

Data 602

Final Project: Firearm Suicide Data 2010-2020

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Abstract

Suicide by firearm remains a major public health issue in the United States. This project investigates whether stronger state-level gun regulation were associated with lower firearm suicide death rates between 2010 and 2020. Mortality data was obtained from the CDC WONDER database and merged with state gun law strength rankings derived from Giffords Law Center scorecards. After cleaning and restructuring the dataset, exploratory data analysis was conducted to examine firearm suicide rates by state, age groups, and time. Statistical analyses included correlation analysis, a one-way ANOVA test, and Tukey's Honestly Significant Difference post hoc test. **Results suggested that there was a moderate negative correlation between gun law strength and firearm suicide death rates, indicating that states with stronger firearm regulations generally experienced lower firearm suicide mortality.** The ANOVA test found statistically significant differences in firearm suicide death rates across gun law categories, while the Tukey HSD test showed that states with weak gun laws had significantly higher rates than states with strong gun laws. Additional findings revealed that firearm suicide rates were highest among older adults and generally increased over time overall. Although the results demonstrate an association between firearm policy strength and firearm suicide mortality, this project does not establish causation and is limited by the use of simplified law score categories and the exclusion of other potentially important socioeconomic and mental health variables.

Introduction

Research Question: Did Stronger Gun Control Laws Result in Less Firearm Suicide Deaths From 2010-2020?

This project investigates the relationship between the strength of state gun regulation laws and firearm suicide death rates in the United States from 2010 through 2020. Specifically, the project analyzes whether states with stronger firearm regulation experienced lower average firearm suicide death rates than states with weaker regulations. In addition to evaluating differences between gun law categories, the analysis also explores how firearm suicide rates varied across age groups and changed over time.

The primary dataset used in this analysis was obtained from the CDC WONDER mortality database and includes firearm suicide deaths categorized by state, year, and age group. State gun law strength rankings were incorporated using Giffords Law Center scorecards and converted into a simplified likert scale representing varying levels of firearm regulation strength.

Variables:

Suicide Deaths by Firearm is the dependent (response) variable and is a quantitative continuous variable measured as deaths per 100,000 population.

Gun Law Strength Score is the independent (explanatory) variable and is an ordinal quantitative variable ranging from 1 (weakest laws) to 5 (strongest laws).

Data Sources

The primary dataset used in this project was obtained from the CDC WONDER Underlying Cause of Death database. The extracted dataset used in this project includes firearm suicide mortality statistics by state, year, and age group from 2010 to 2020.

State gun law strength information was derived from Giffords Law Center state scorecards. States were assigned numerical categories ranging from 1 (weakest firearm regulations) to 5 (strongest firearm regulations).

Data Wrangling

Data Cleaning:

- Removed non-data rows and explanatory footnotes by dropping rows missing valid state identifiers (State Code).
- Renamed columns to improve readability and consistency throughout the analysis.
- Removed unnecessary columns that were not required for analysis, including
- Replaced nonnumeric values labeled "Unreliable" into missing values (NaN) for easier handling during analysis.
- Converted data types to consistent formats.
- Grouped and aggregated data by: State, Year, Age group and Gun law strength category.
- Verified that cleaned datasets retained only variables relevant to the research question and statistical analyses.

Data Merging:

- Created a secondary dataframe containing state-level gun law strength scores derived from Giffords Law Center rankings.
- Assigned each state a simplified numerical gun law score ranging from:
 - 1 = weakest firearm regulations
 - 5 = strongest firearm regulations
- Merged the CDC firearm suicide dataset with the gun law score dataframe using state names as the common merge key.

Missing Values

The original CDC dataset contained several firearm suicide death rate values labeled as “Unreliable.” These values typically occurred in age groups or states with smaller population sizes where mortality estimates may not have been statistically stable. During preprocessing, these entries were converted into missing values (NaN) using numeric coercion.

Exploratory Data Analysis



Mean Firearm Death Rate by State (Top 10)

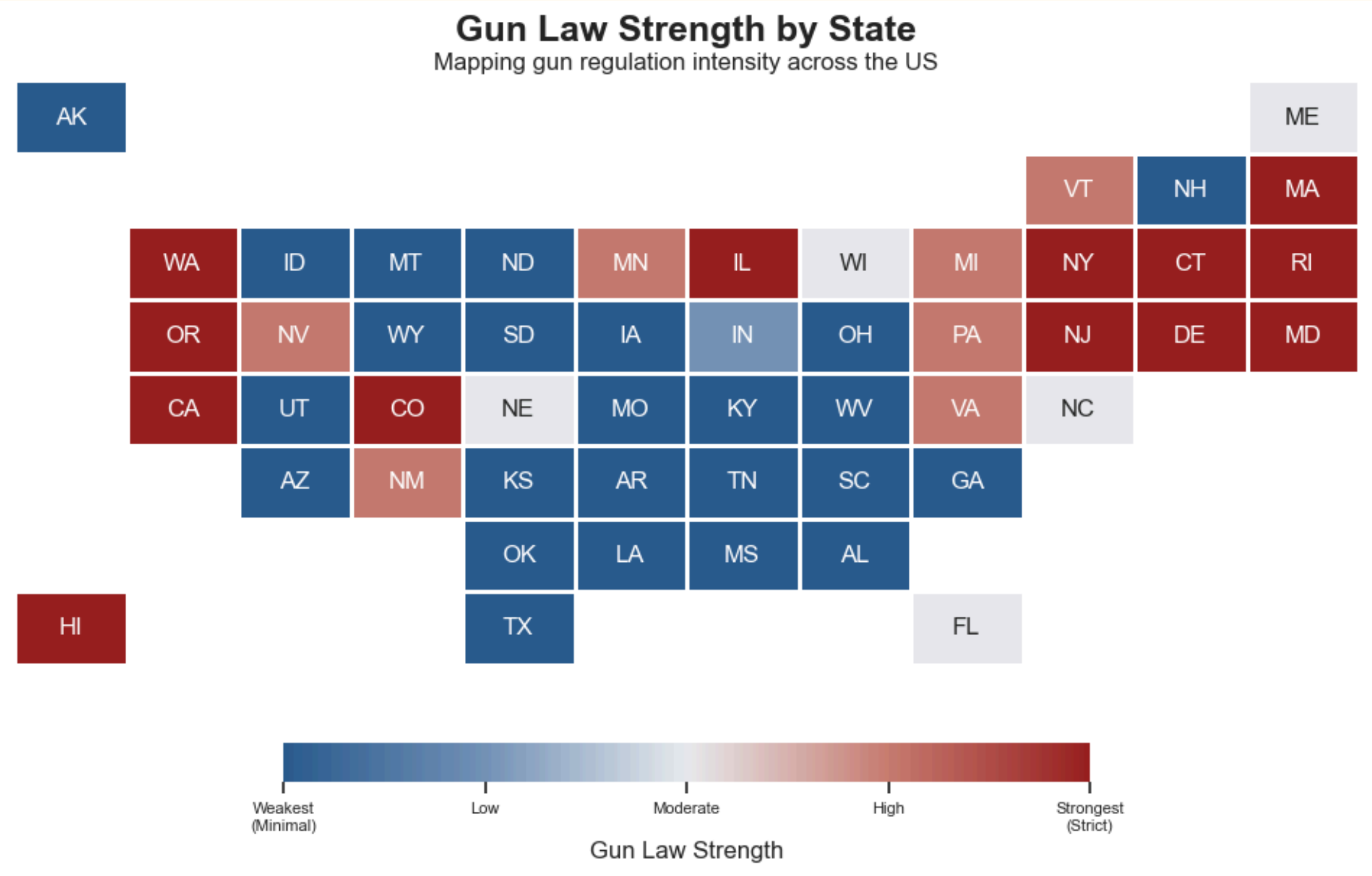
States with weaker gun law environments, particularly western states, consistently showed the highest average firearm suicide death rates from 2010–2020.

Table 1: Mean Firearm Death Rate by State (Top 10)

State	Death Rate	State	Death Rate
Wyoming	29.29	Idaho	16.40
Alaska	24.49	New Mexico	15.44
North Dakota	21.45	West Virginia	15.39
Montana	20.24	Nevada	15.12
South Dakota	19.40	Oklahoma	15.11

Gun Law Strength by State

The heatmap highlights geographic differences in firearm regulation strength, with stronger gun laws concentrated primarily in coastal and northeastern states.



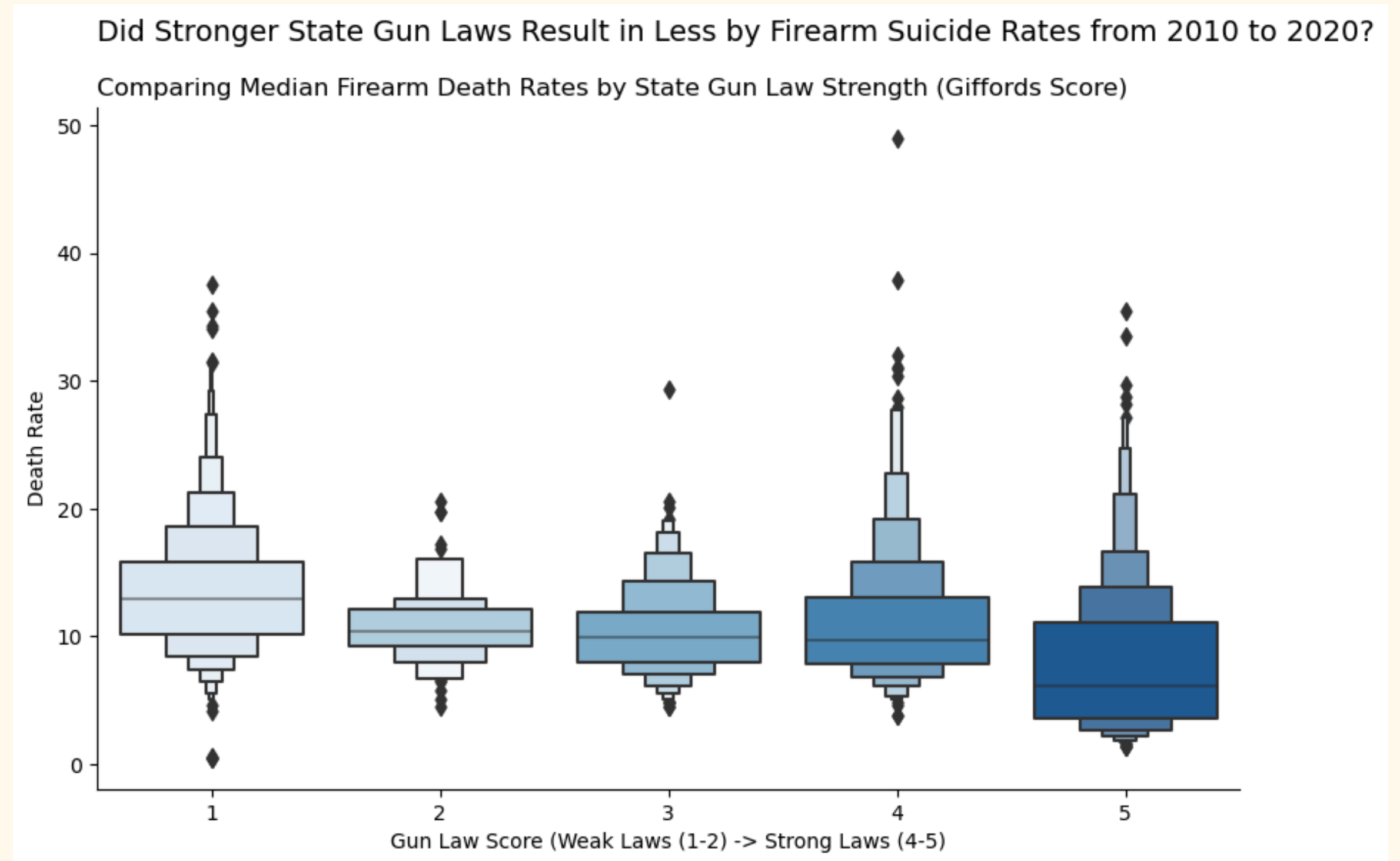
Death Rates by Gun Law Strength: Averages

States with stronger gun laws demonstrated lower average firearm suicide death rates compared to states with weaker firearm regulations.

Law Score	Average Rate	Median Rate	Std. Deviation	Minimum	Maximum
1	13.46	12.90	4.78	0.50	37.50
2	10.77	10.40	2.99	4.50	20.60
3	10.39	9.90	3.41	4.50	29.30
4	11.11	9.70	5.10	3.80	49.00
5	7.87	6.15	5.53	1.40	35.50

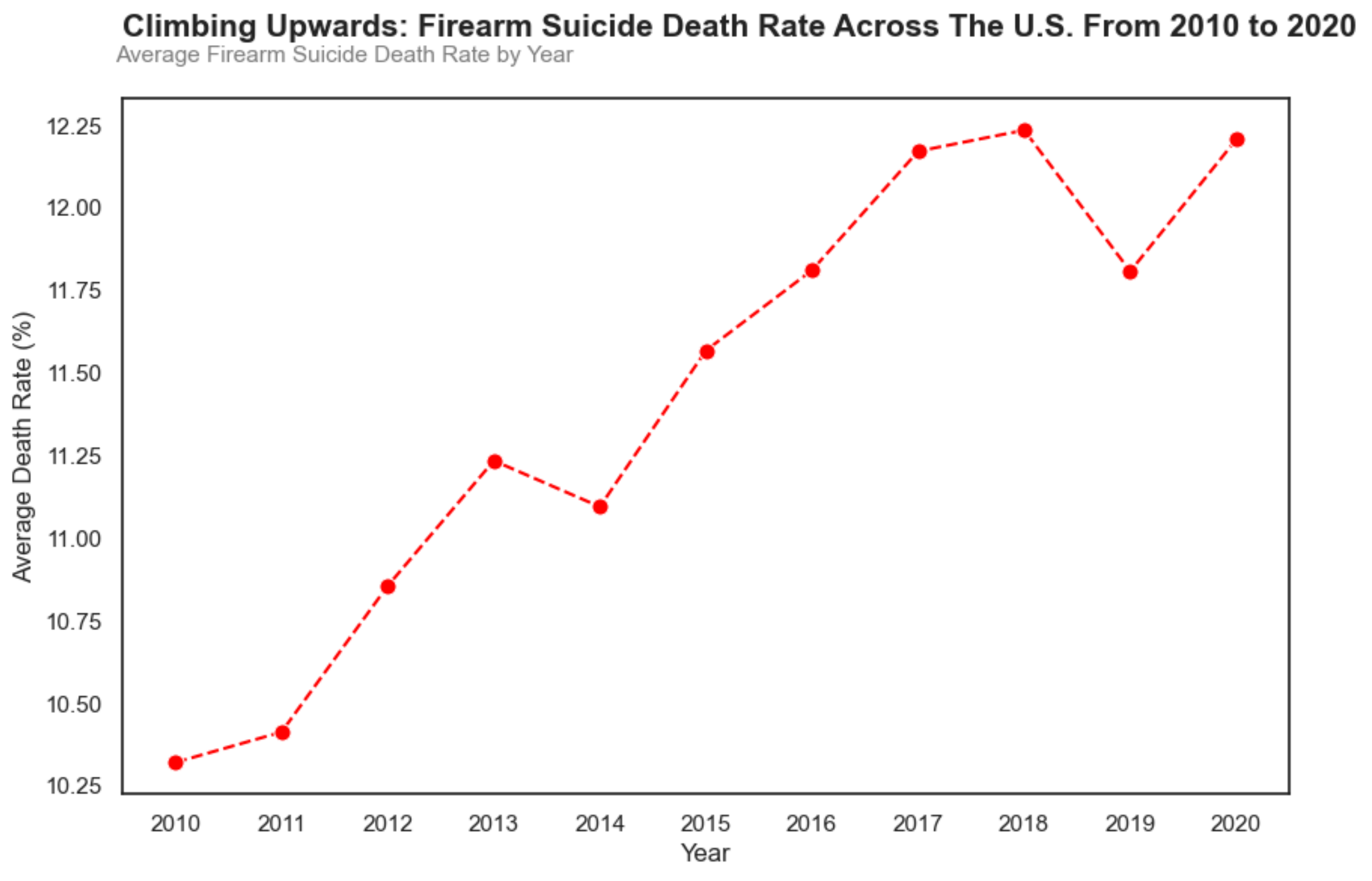
Death Rates by Gun Law Strength: Medians

The boxen plot highlights that the median and spread of suicide death rates under strong gun law regulation (score 5) are lower and smaller than those of death rates under weaker regulation (score 1-4).



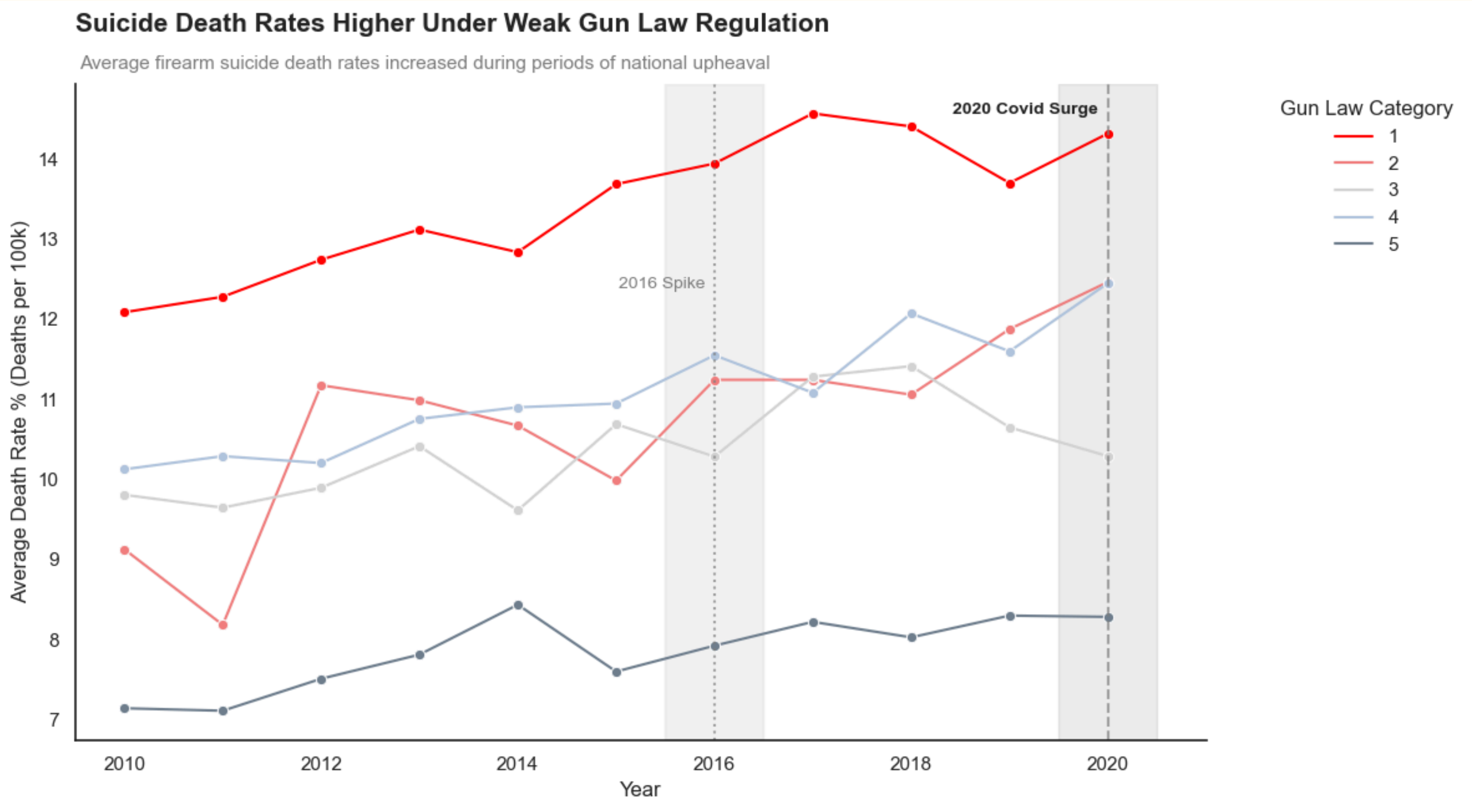
Firearm Suicide Rates Over Time

Firearm suicide death rates steadily increased throughout the decade, with noticeable increases occurring around 2016 and during the COVID-19 pandemic in 2020.



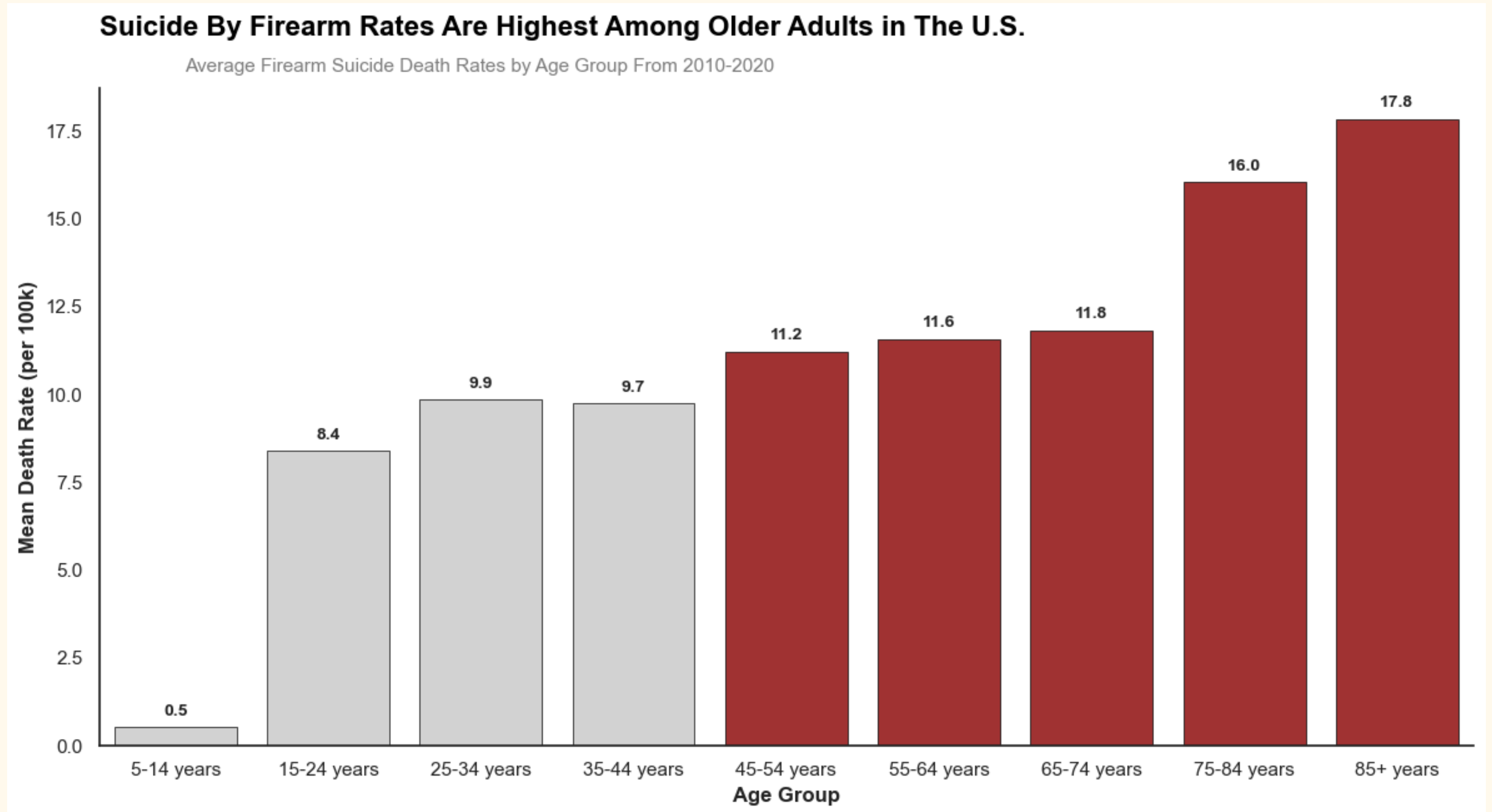
Death Rates by Year and Gun Law Strength

States with weaker firearm regulations consistently maintained higher firearm suicide death rates across nearly all years analyzed.



Firearm Suicide Rates by Age Group

Older adult populations experienced disproportionately higher firearm suicide death rates, while rates among children and adolescents remained comparatively low.



Statistical Analysis

Correlation Analysis

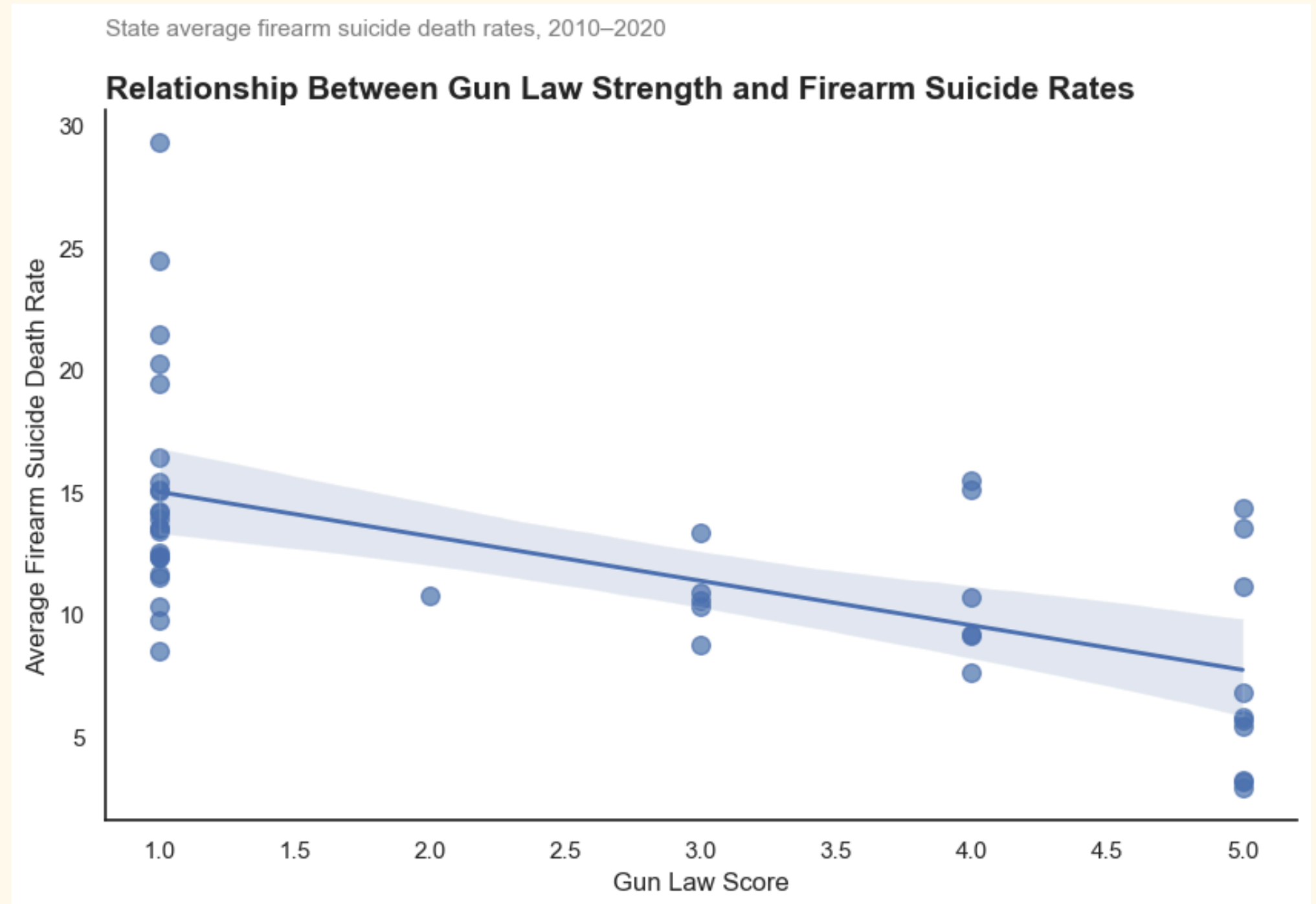
Correlation analysis found that there is a moderate-to-strong negative correlation (-0.595), it appears that stronger gun laws tend to be associated with lower firearm suicide death rates.

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CORRELATION ANALYSIS SUMMARY
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Variables: Law_Score vs. Avg_Death_Rate
Pearson r: -0.595
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Linear Regression

A simple linear regression found a statistically significant negative relationship between gun law strength and firearm suicide death rates ($r = -0.595$, $p < 0.001$). The model estimated that each one-point increase in gun law strength score was associated with approximately 1.82 fewer firearm suicide deaths per 100,000 people.

However, since our gun law strength score can only be one of five values, our data is categorical and a linear model is not the best option.



Slope: -1.8194126916443545
R-squared: 0.3539589120589069
P-value: $1.2976296950465467e-05$

ANOVA TEST: ASSUMPTIONS

Prior to conducting the ANOVA test, Levene’s test was used to check that the assumption of homogeneity of variances across gun law strength groups is met. The dependent variable, average firearm suicide death rate, was continuous, and observations were treated as independent at the state level, thus meeting the two assumptions to run the Leven test.

Metric	Value
W-Statistic	1.3278
p-value	0.2757
Assumption status	Equal Variance Met

ANOVA: ONE-WAY TEST AND TUKEY TEST

The ANOVA test result suggests that states with different levels of gun law strength experienced significantly different average firearm suicide death rates from 2010-2020.

Metric	Value
F-Statistic	9.7831
p-value	0.0003
Result	Significant

A Tukey HSD post-hoc test found that states classified as having weak gun laws had significantly higher firearm suicide death rates than states classified as having strong gun laws ($p < 0.001$). These findings suggest that the largest differences in firearm suicide mortality occurred between states with the strongest and weakest firearm policy environments from 2010-2020.

Multiple Comparison of Means – Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
Moderate	Strong	-2.0675	0.6369	-7.5814	3.4464	False
Moderate	Weak	4.1055	0.1538	-1.1668	9.3778	False
Strong	Weak	6.173	0.0002	2.7275	9.6186	True

Limitations

Several limitations should be considered when interpreting the findings of this project.

- First, the analysis identifies associations between gun law strength and firearm suicide death rates but does not establish causation. Other factors may influence firearm suicide mortality, such as socioeconomic conditions, mental health access, firearm ownership rates, and varying levels of access to healthcare.
- Second, the gun law strength scores used in this study were simplified from Giffords Law Center scorecards and manually categorized into broad numerical groups. These categories may not fully capture the complex and changing nature of firearm legislation over time. Additionally, some gun law categories contained relatively small numbers of states, which limited statistical balance across groups.
- Another limitation involves the use of state-level aggregated data. Statewide averages may conceal important local or regional differences within states. Also, some firearm death rate observations were labeled as unreliable by the CDC and were excluded from portions of the analysis.

Conclusion

Did Stronger Gun Control Laws Result in Less Firearm Suicide Deaths From 2010-2020?

- This project investigated the relationship between state gun law strength and firearm suicide death rates in the United States between 2010 and 2020. Through statistical analysis, the findings revealed that **states with stronger firearm regulations generally experienced lower firearm suicide death rates than states with weaker regulations.**
- **A Tuskey Test further demonstrated significant differences between strong-law and weak-law states.**
- The analysis also identified important demographic and temporal trends. Firearm suicide death rates were highest among older adults and generally increased nationwide from 2010 to 2020. These findings suggest that firearm suicide remains an important and evolving public health issue across the United States.
- However, the results do not prove causation. Future research that incorporates additional variables such as firearm ownership rates, mental health resources, socioeconomic indicators, and changes in firearm legislation over time could provide further insight into the findings of this project.

Thank
You

